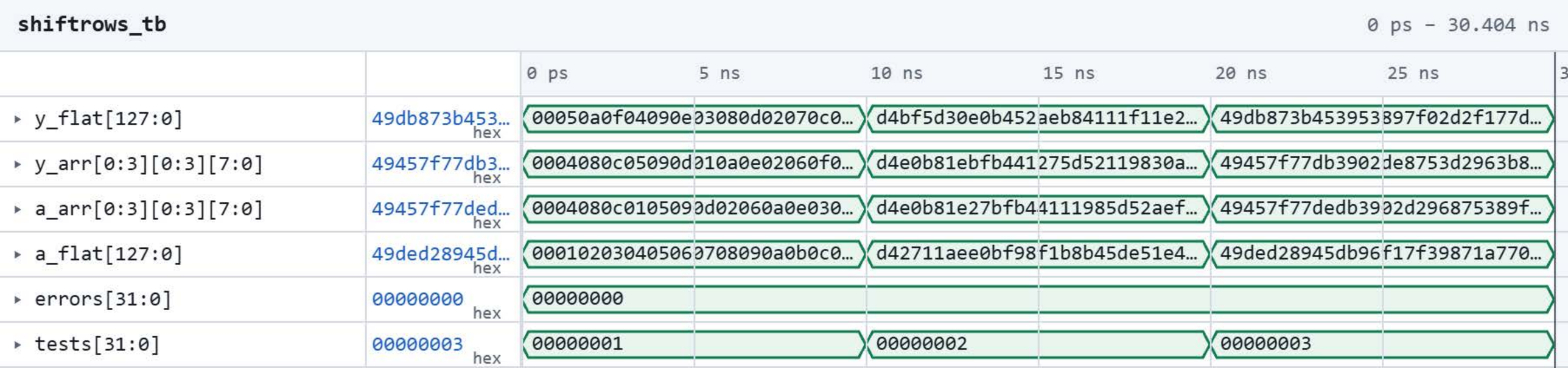
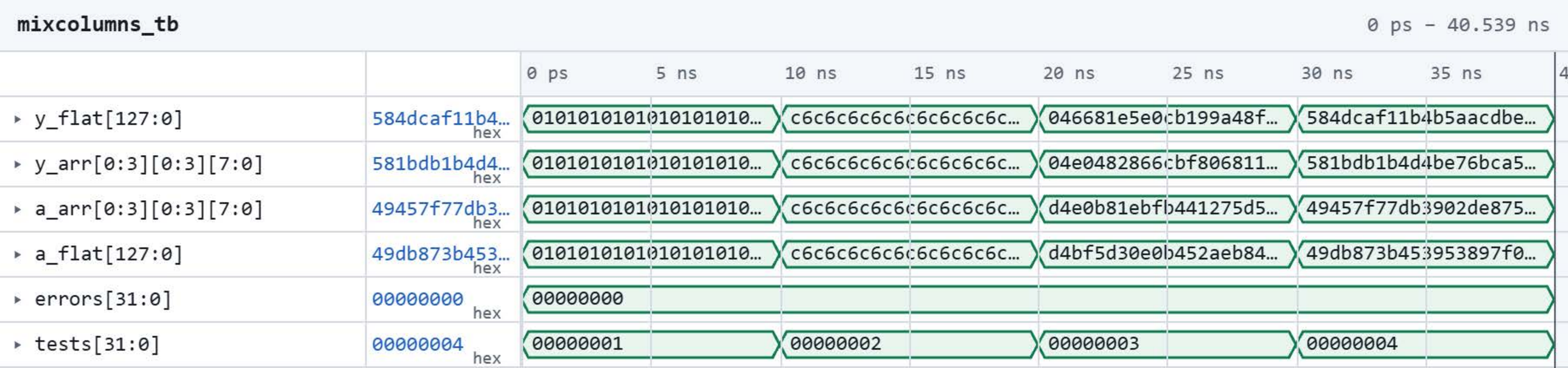
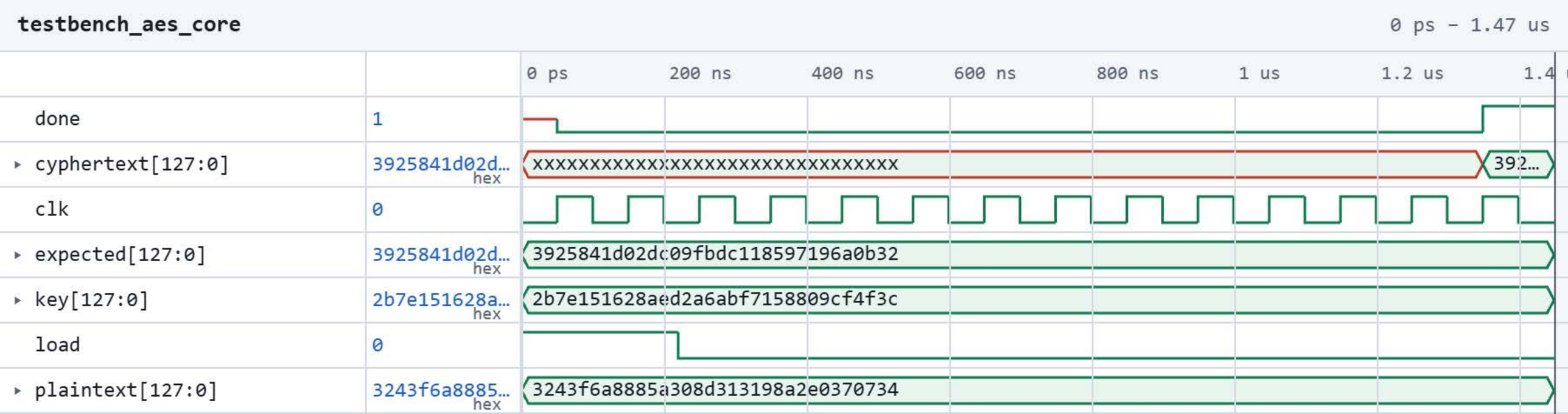
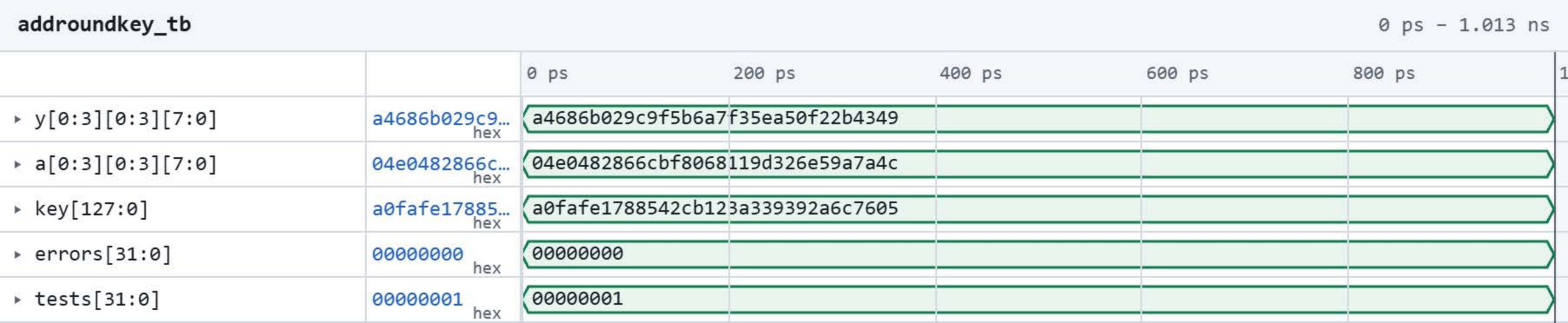

shiftrows_tb
0 ps – 30.404 ns
0 ps
5 ns
10 ns
15 ns
20 ns
25 ns
y_flat[127:0]
49db873b453…
hex
00050a0f04090e03080d02070c0…
d4bf5d30e0b452aeb84111f11e2…
49db873b453953897f02d2f177d…
y_arr[0:3][0:3][7:0]
49457f77db3…
hex
0004080c05090d010a0e02060f0…
d4e0b81ebfb441275d52119830a…
49457f77db3902de8753d2963b8…
a_arr[0:3][0:3][7:0]
49457f77ded…
hex
0004080c0105090d02060a0e030…
d4e0b81e27bfb44111985d52aef…
49457f77dedb3902d296875389f…
a_flat[127:0]
49ded28945d…
hex
000102030405060708090a0b0c0…
d42711aee0bf98f1b8b45de51e4…
49ded28945db96f17f39871a770…
errors[31:0]
00000000
hex
00000000
tests[31:0]
00000003
hex
00000001
00000002
00000003

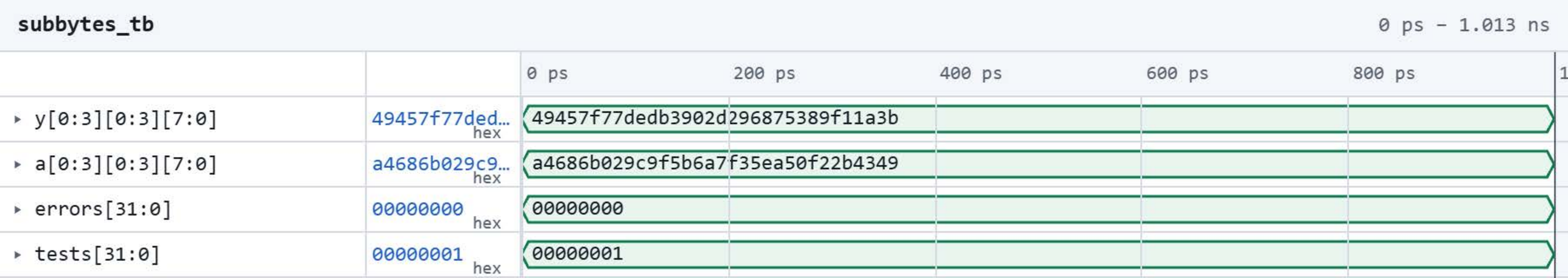

mixcolumns_tb
0 ps – 40.539 ns
0 ps
5 ns
10 ns
15 ns
20 ns
25 ns
30 ns
35 ns
y_flat[127:0]
584dcaf11b4…
hex
01010101010101010…
c6c6c6c6c6c6c6c6c6c…
046681e5e0cb199a48f…
584dcaf11b4b5aacdbe…
y_arr[0:3][0:3][7:0]
581bdb1b4d4…
hex
01010101010101010…
c6c6c6c6c6c6c6c6c6c…
04e0482866cbf806811…
581bdb1b4d4be76bca5…
a_arr[0:3][0:3][7:0]
49457f77db3…
hex
01010101010101010…
c6c6c6c6c6c6c6c6c6c…
d4e0b81ebfb441275d5…
49457f77db3902de875…
a_flat[127:0]
49db873b453…
hex
01010101010101010…
c6c6c6c6c6c6c6c6c6c…
d4bf5d30e0b452aeb84…
49db873b453953897f0…
errors[31:0]
00000000
hex
00000000
tests[31:0]
00000004
hex
00000001
00000002
00000003
00000004

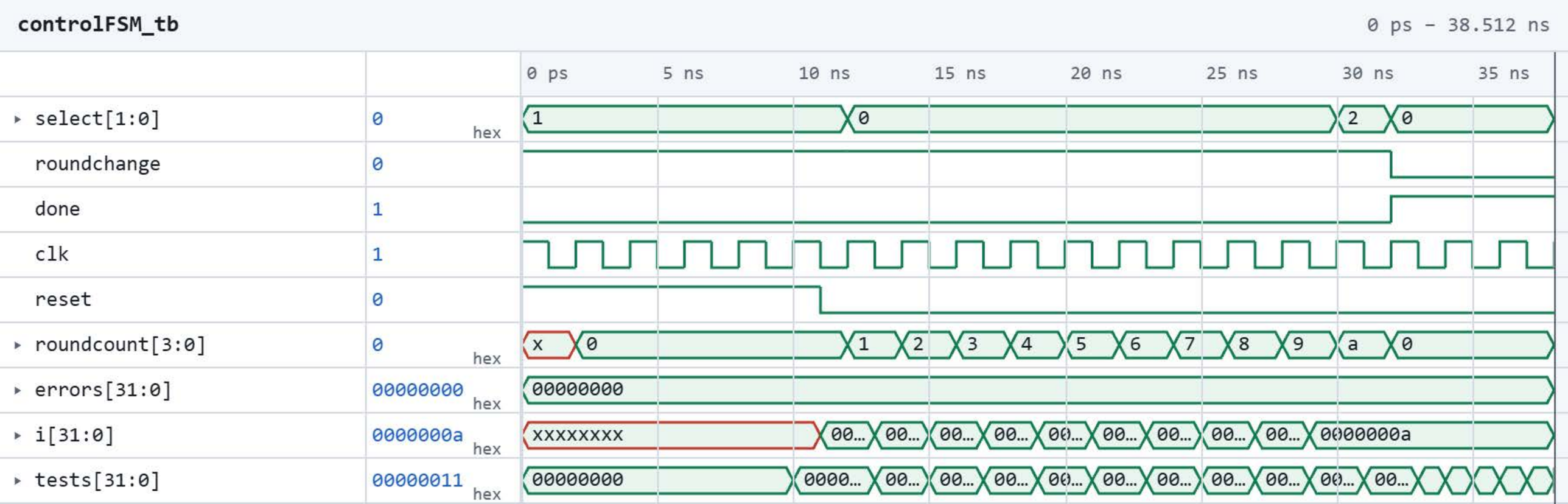

testbench_aes_core
0 ps – 1.47 us
0 ps
200 ns
400 ns
600 ns
800 ns
1 us
1.2 us
done
1
cyphertext[127:0]
3925841d02d…
hex
clk
0
expected[127:0]
3925841d02d…
hex
3925841d02dc09fbdc118597196a0b32
key[127:0]
2b7e151628a…
hex
2b7e151628aed2a6abf7158809cf4f3c
load
0
plaintext[127:0]
3243f6a8885…
hex
3243f6a8885a308d313198a2e0370734

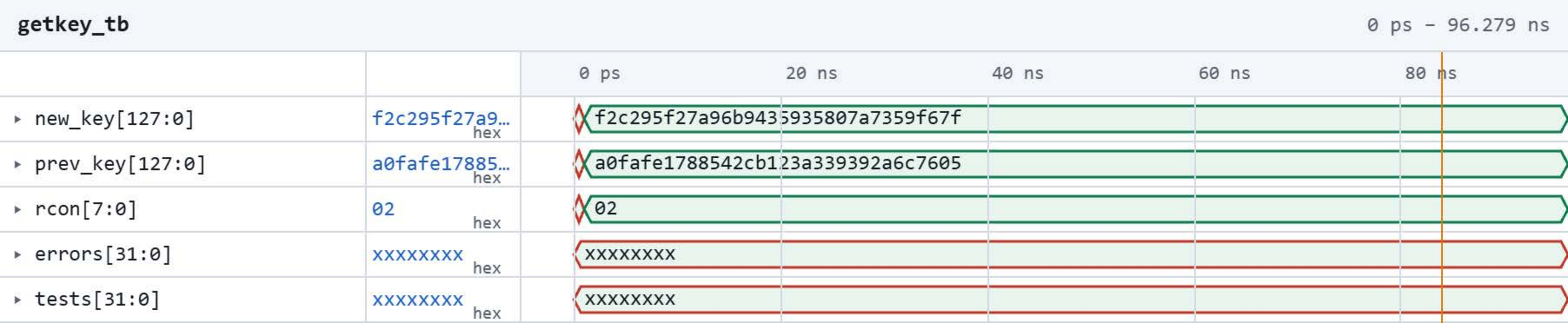
aes_core_trace_tb 170.216 ns – 1.549 us

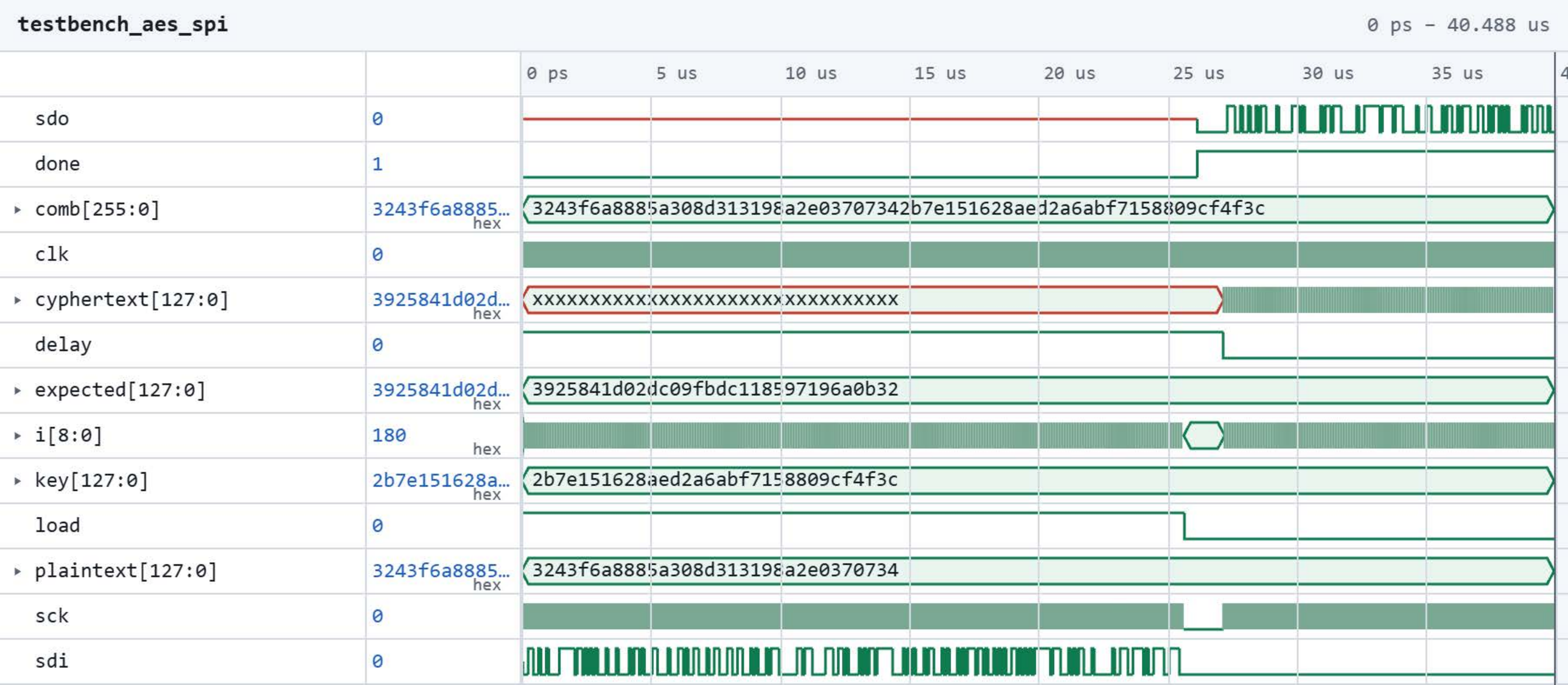
		200 ns	400 ns	600 ns	800 ns	1 us	1.2 us	1.4 us
▸ key_probe[127:0]	xxeadde6xx0… hex	xx…	a0fa…	f2c2…	3d80…	ef44…	d4d1…	6d88…	4e54…	ead2…	ac77…	d014…	xxea…
▸ state_probe[127:0]	3925841d02d… hex	193de3bea…	a49c…	aa8f…	486c…	e092…	f100…	260e…	5a41…	ea83…	eb40…	3925…
done	1												
▸ cyphertext[127:0]	3925841d02d… hex	xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx											
clk	1												
▸ expected[127:0]	3925841d02d… hex	3925841d02dc09fbdc118597196a0b32											
▸ key[127:0]	2b7e151628a… hex	2b7e151628aed2a6abf7158809cf4f3c											
▸ key_last[127:0]	xxxxxxxxxxx… hex	xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx											
load	0												
▸ plaintext[127:0]	3243f6a8885… hex	3243f6a8885a308d313198a2e0370734											
▸ state_last[127:0]	3925841d02d… hex	193de3bea0…	a49c…	aa8f…	486c…	e092…	f100…	260e…	5a41…	ea83…	eb40…	3925…
▸ cycle[31:0]	0000000e hex	000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…
▸ done_x[31:0]	00000000 hex	00000000											
▸ key_good[31:0]	ffffffff hex	ffffffff											
▸ key_x[31:0]	00000000 hex	00000000											
▸ mismatches[31:0]	00000000 hex	00000000											
▸ state_good[31:0]	00000033 hex	00000002	0000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…	0000…
▸ state_x[31:0]	00000000 hex	00000000											

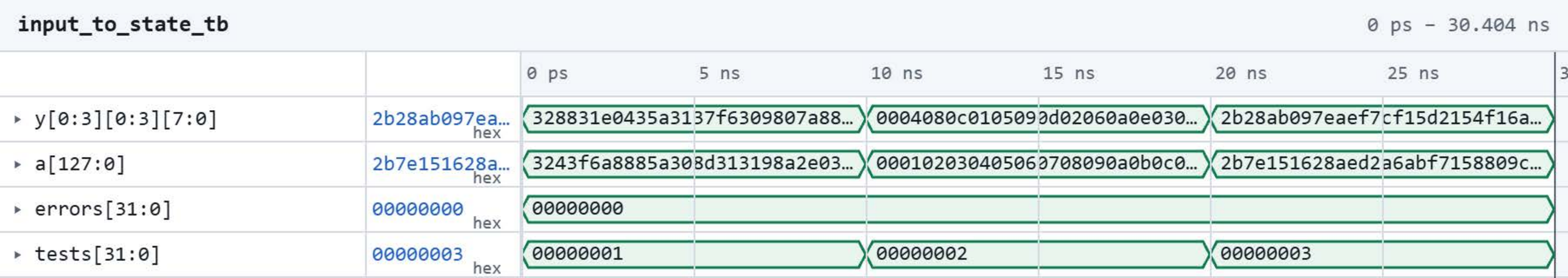

addroundkey_tb
0 ps – 1.013 ns
0 ps
200 ps
400 ps
600 ps
800 ps
y[0:3][0:3][7:0]
a4686b029c9…
hex
a4686b029c9f5b6a7f35ea50f22b4349
a[0:3][0:3][7:0]
04e0482866c…
hex
04e0482866cbf8068119d326e59a7a4c
key[127:0]
a0fafe17885…
hex
a0fafe1788542cb123a339392a6c7605
errors[31:0]
00000000
hex
00000000
tests[31:0]
00000001
hex
00000001

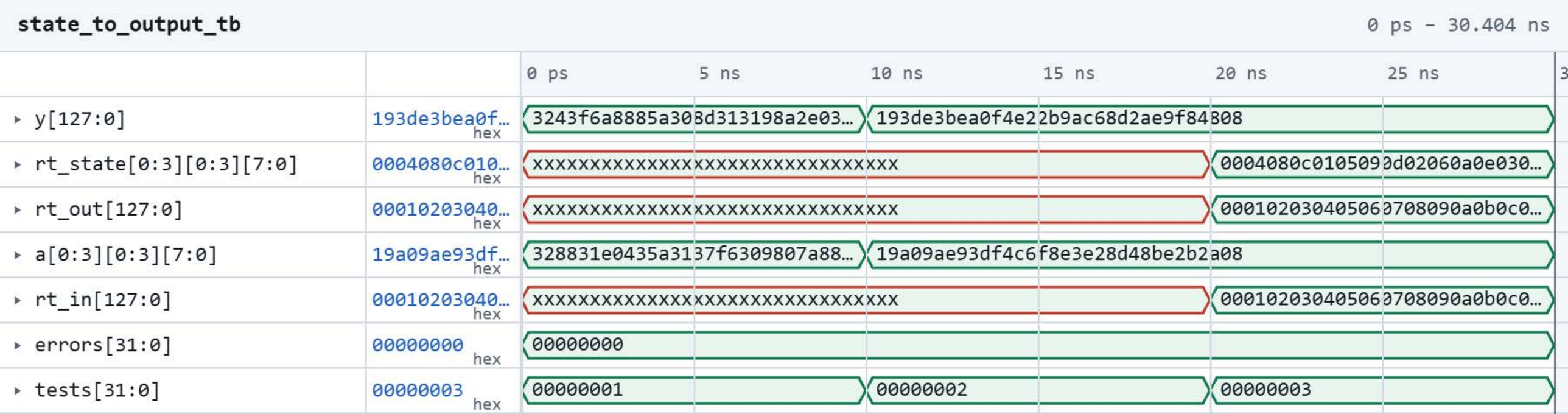

subbytes_tb
0 ps – 1.013 ns
0 ps
200 ps
400 ps
600 ps
800 ps
y[0:3][0:3][7:0]
49457f77ded…
hex
49457f77dedb3902d296875389f11a3b
a[0:3][0:3][7:0]
a4686b029c9…
hex
a4686b029c9f5b6a7f35ea50f22b4349
errors[31:0]
00000000
hex
00000000
tests[31:0]
00000001
hex
00000001

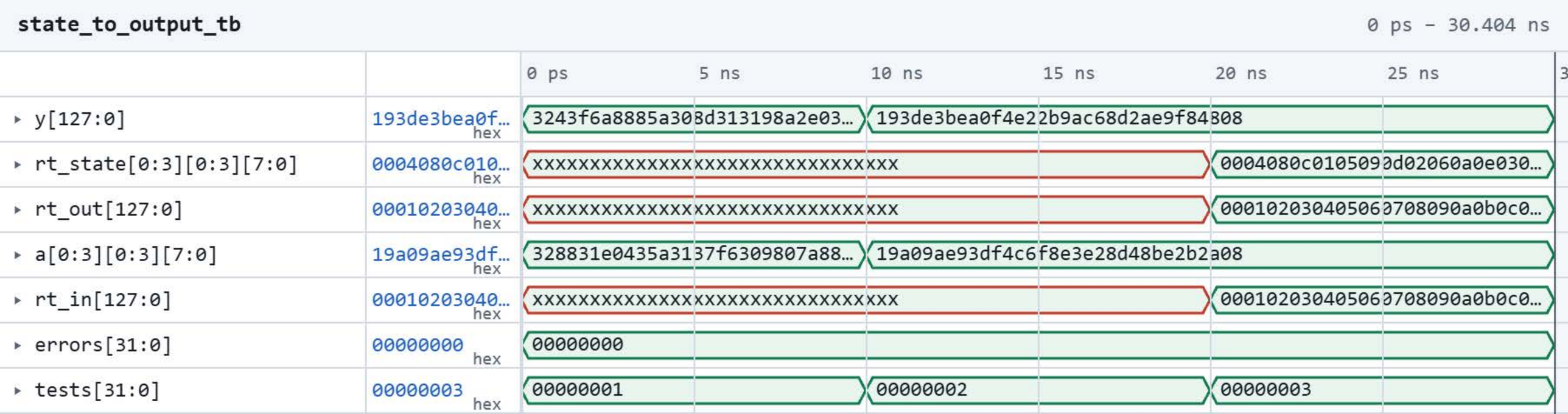

controlFSM_tb
0 ps – 38.512 ns
0 ps
5 ns
10 ns
15 ns
20 ns
25 ns
30 ns
35 ns
select[1:0]
0
hex
roundchange
0
done
1
clk
1
reset
0
roundcount[3:0]
0
hex
errors[31:0]
00000000
hex
i[31:0]
0000000a
hex
tests[31:0]
00000011
hex


getkey_tb
0 ps – 96.279 ns
0 ps
20 ns
40 ns
60 ns
80 ns
new_key[127:0]
f2c295f27a9…
hex
f2c295f27a96b9435935807a7359f67f
prev_key[127:0]
a0fafe17885…
hex
a0fafe1788542cb123a339392a6c7605
rcon[7:0]
02
hex
02
errors[31:0]
xxxxxxxx
hex
xxxxxxxx
tests[31:0]
xxxxxxxx
hex
xxxxxxxx


testbench_aes_spi
0 ps – 40.488 us
0 ps
5 us
10 us
15 us
20 us
25 us
30 us
35 us
sdo
0
done
1
comb[255:0]
3243f6a8885...
hex
3243f6a8885a308d313198a2e03707342b7e151628aed2a6abf7158809cf4f3c
clk
0
cyphertext[127:0]
3925841d02d...
hex
delay
0
expected[127:0]
3925841d02d...
hex
3925841d02dc09fbdc11897196a0b32
i[8:0]
180
hex
key[127:0]
2b7e151628a...
hex
2b7e151628aed2a6abf7158809cf4f3c
load
0
plaintext[127:0]
3243f6a8885...
hex
3243f6a8885a308d313198a2e0370734
sck
0
sdi
0


input_to_state_tb
0 ps – 30.404 ns
0 ps
5 ns
10 ns
15 ns
20 ns
25 ns
y[0:3][0:3][7:0]
2b28ab097ea…
hex
328831e0435a3137f6309807a88…
0004080c0105090d02060a0e030…
2b28ab097eaef7cf15d2154f16a…
a[127:0]
2b7e151628a…
hex
3243f6a8885a308d313198a2e03…
000102030405060708090a0b0c0…
2b7e151628aed2a6abf7158809c…
errors[31:0]
00000000
hex
00000000
tests[31:0]
00000003
hex
00000001
00000002
00000003


state_to_output_tb
0 ps – 30.404 ns
0 ps
5 ns
10 ns
15 ns
20 ns
25 ns
y[127:0]
193de3bea0f…
hex
3243f6a8885a308d313198a2e03…
193de3bea0f4e22b9ac68d2ae9f84808
rt_state[0:3][0:3][7:0]
0004080c010…
hex
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
0004080c0105090d02060a0e030…
rt_out[127:0]
00010203040…
hex
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
000102030405060708090a0b0c0…
a[0:3][0:3][7:0]
19a09ae93df…
hex
328831e0435a3137f6309807a88…
19a09ae93df4c6f8e3e28d48be2b2a08
rt_in[127:0]
00010203040…
hex
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
000102030405060708090a0b0c0…
errors[31:0]
00000000
hex
00000000
tests[31:0]
00000003
hex
00000001
00000002
00000003


state_to_output_tb
0 ps – 30.404 ns
0 ps
5 ns
10 ns
15 ns
20 ns
25 ns
y[127:0]
193de3bea0f…
hex
3243f6a8885a308d313198a2e03…
193de3bea0f4e22b9ac68d2ae9f84808
rt_state[0:3][0:3][7:0]
0004080c010…
hex
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
0004080c0105090d02060a0e030…
rt_out[127:0]
00010203040…
hex
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
000102030405060708090a0b0c0…
a[0:3][0:3][7:0]
19a09ae93df…
hex
328831e0435a3137f6309807a88…
19a09ae93df4c6f8e3e28d48be2b2a08
rt_in[127:0]
00010203040…
hex
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
000102030405060708090a0b0c0…
errors[31:0]
00000000
hex
00000000
tests[31:0]
00000003
hex
00000001
00000002
00000003